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Adaptation Needs in Robotic Systems: Assessing Behavior Trees and Their Enhancements

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The study asks three questions

- ❖ RQ1. What drives a robotic system to adapt?
- ❖ RQ2. Where do classical Behavior Trees (BTs) help - and where do they fall short?
- ❖ RQ3. To what extent do BT enhancements address adaptation needs, and what limitations remain or emerge?

The world changes after deployment:



Sensor lost?

Grasp varies?

Task changes?



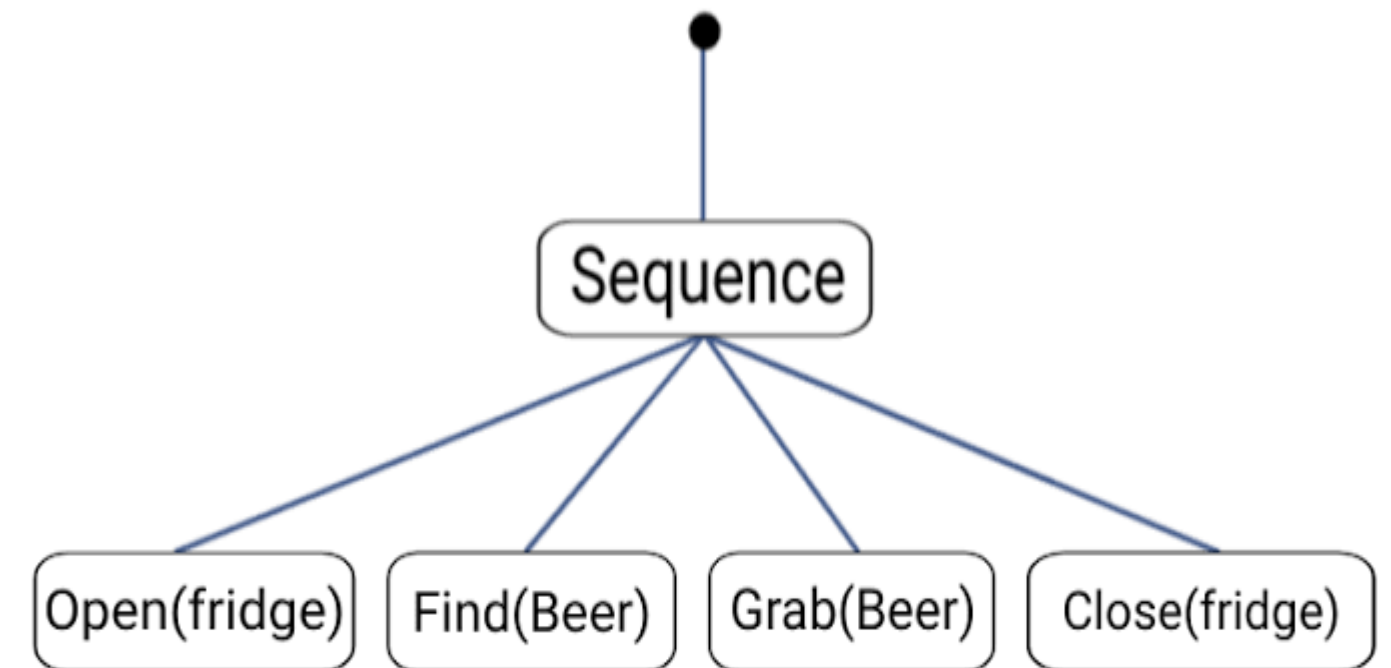
Adaptation needs in robotic systems



Classical Behavior Trees (BTs)

BTs are hierarchical control structures to orchestrate the behavior of autonomous agents.

- Modularity
- Interpretability
- Reactivity
- Reusability
- Maintainability
- Extensibility
- Decomposability
- Flexibility



Are BTs sufficient to meet the adaptation needs of modern robotics?

Where classical BTs fall short

Knowledge	Incompleteness	R ~
System	Component failure	R ~ F ×
Mission	Uncertain specification	M ~ R ~ F ~
Environment	Execution context changes	R ~ F ~
Environment	Non-human agents	R ×
Environment	Human agents	R ~ F ~

Main BT limitations

Rigid/static control flow

Limited handling of uncertainty and changing context

Limited planning, intent, and goal representation

Difficulty representing complex behaviors

Limited task recovery / resumption

Limited representation of quality requirements

Limited support for continuous-valued states/actions

~ Partial / implicit support × No support

R: Reactivity M: Modularity F: Flexibility

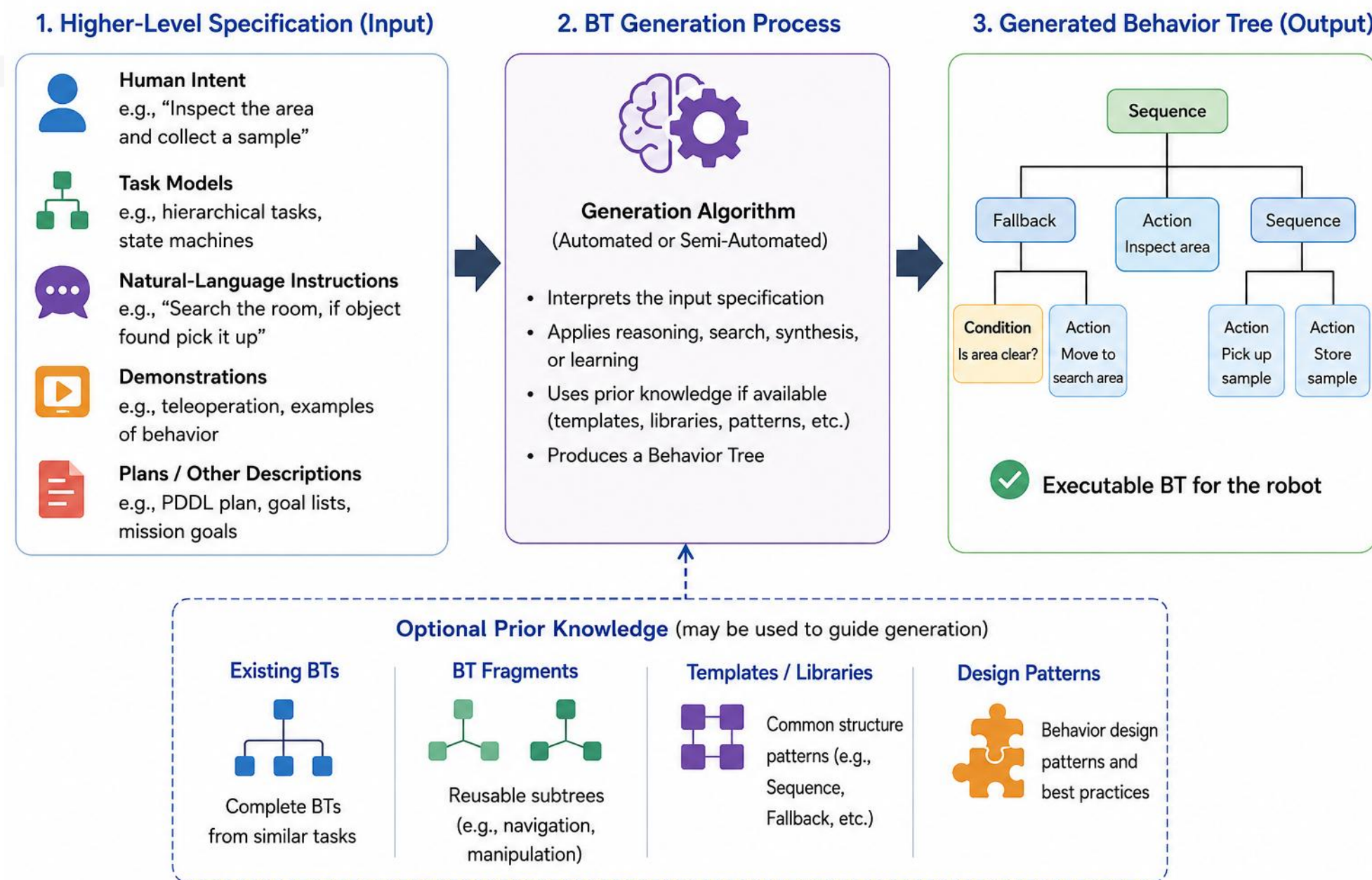
Reactivity & Flexibility are partial across almost all complex scenarios.

Four Families of Enhancements

How does the scientific literature extend
BTs to overcome limitations?

BT Generation

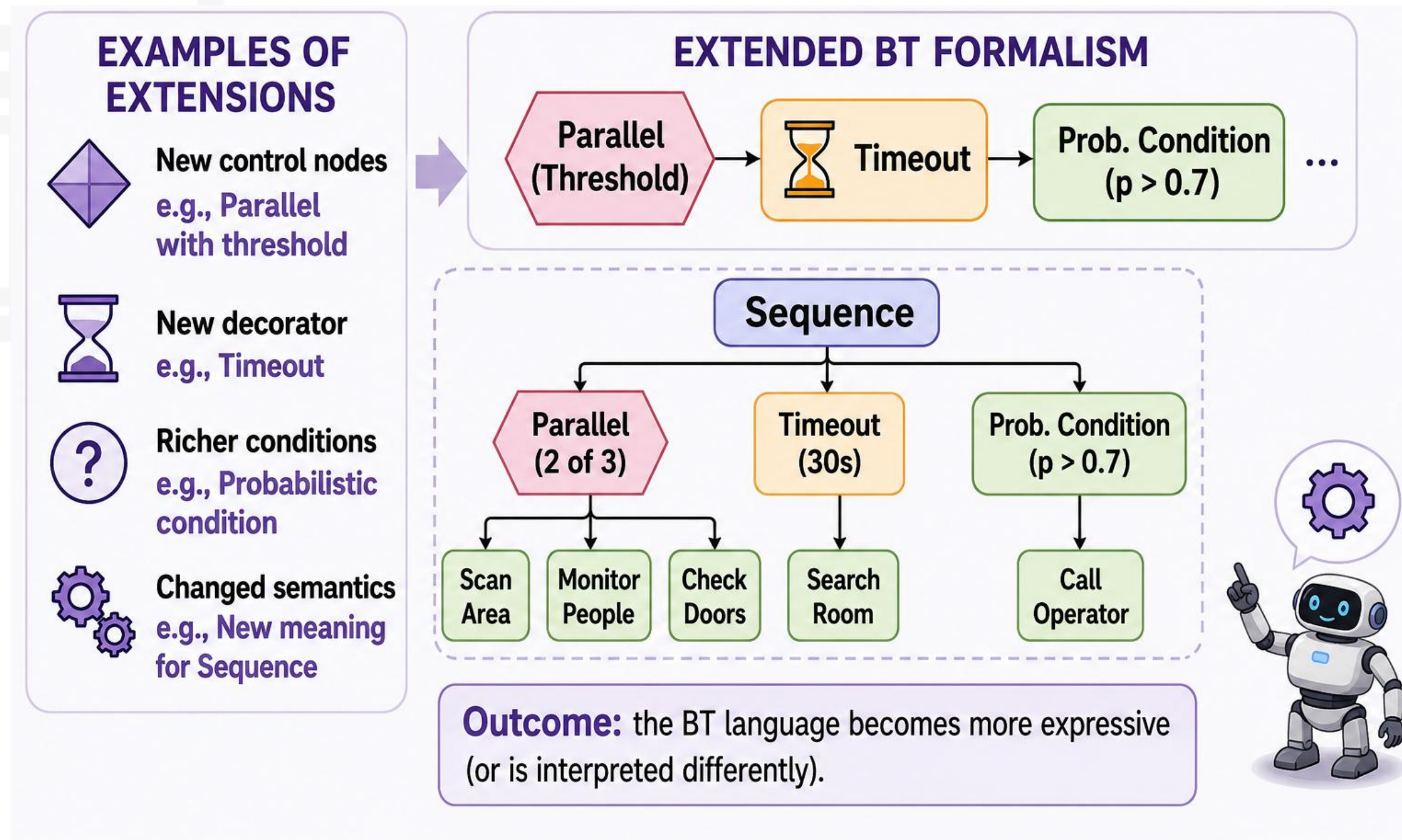
BT Generation refers to an automated or semi-automated processes that produce a BT from a higher-level specification.



No initial BT is taken as the behavior to be modified.

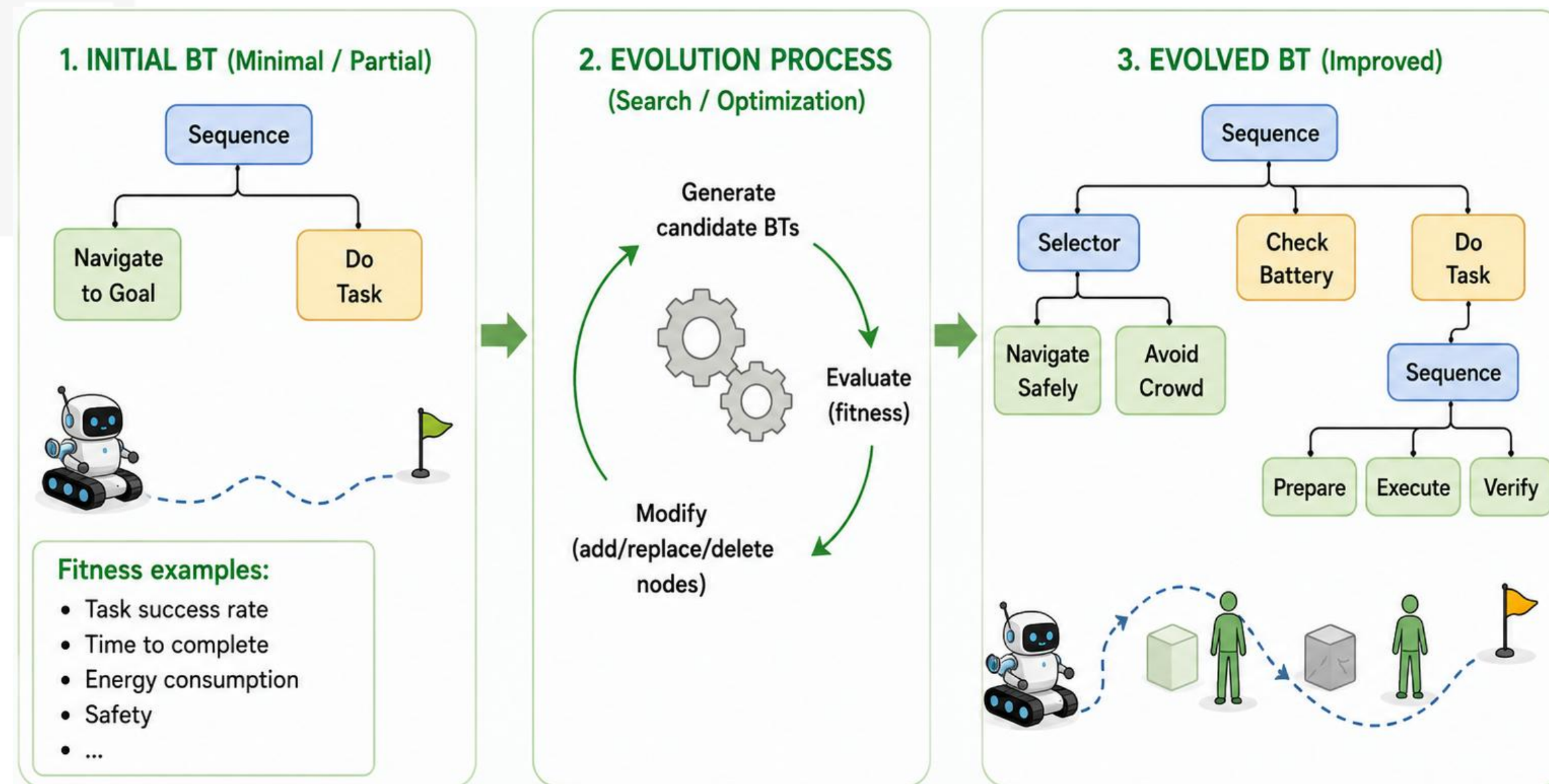
BT Extension

BT Extension refers to processes that modify the BT formalism itself.



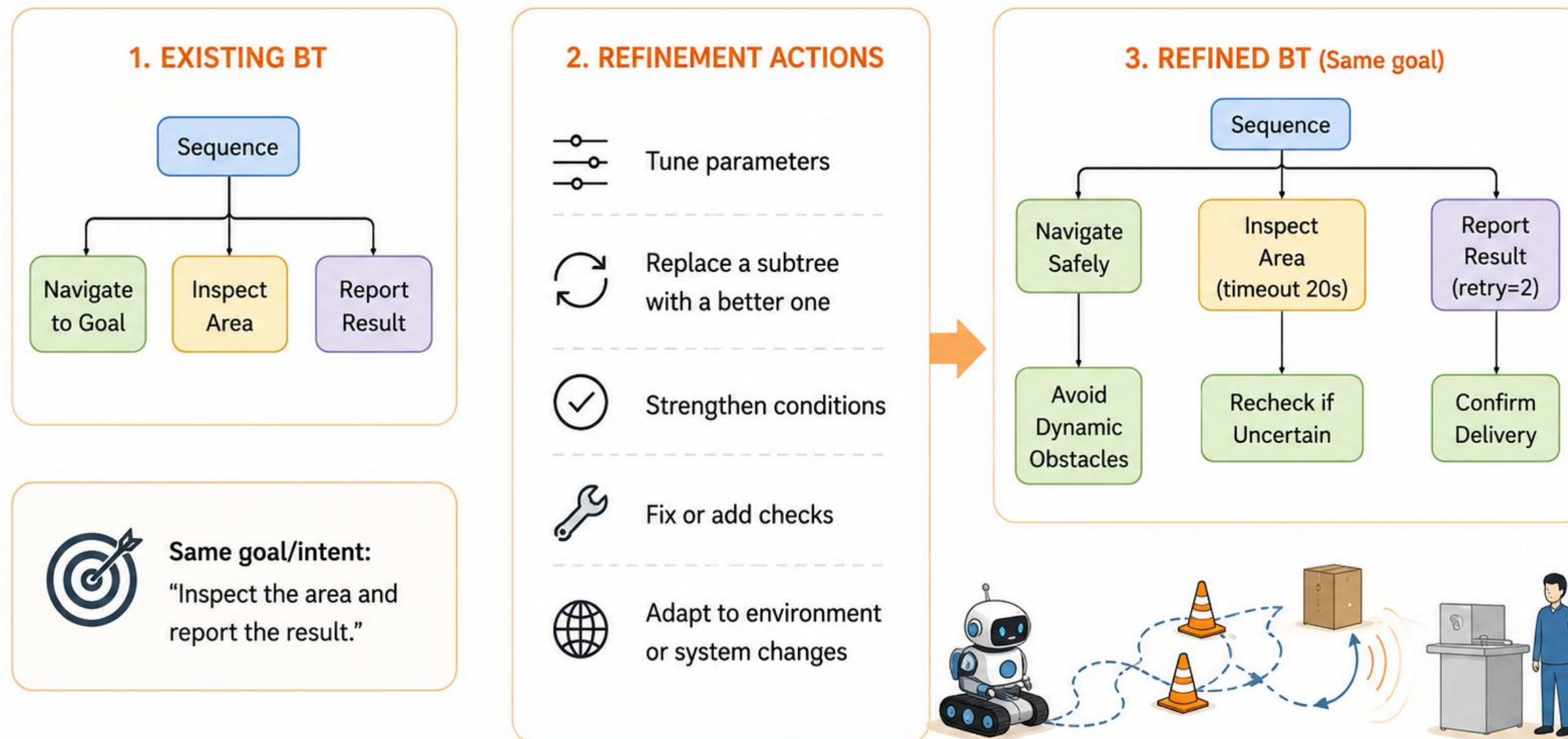
BT Evolution

BT Evolution refers to processes that start from an initial, often minimal or partial, BT and progressively modify its structure to obtain a tree that encodes more complex or improved behavior.



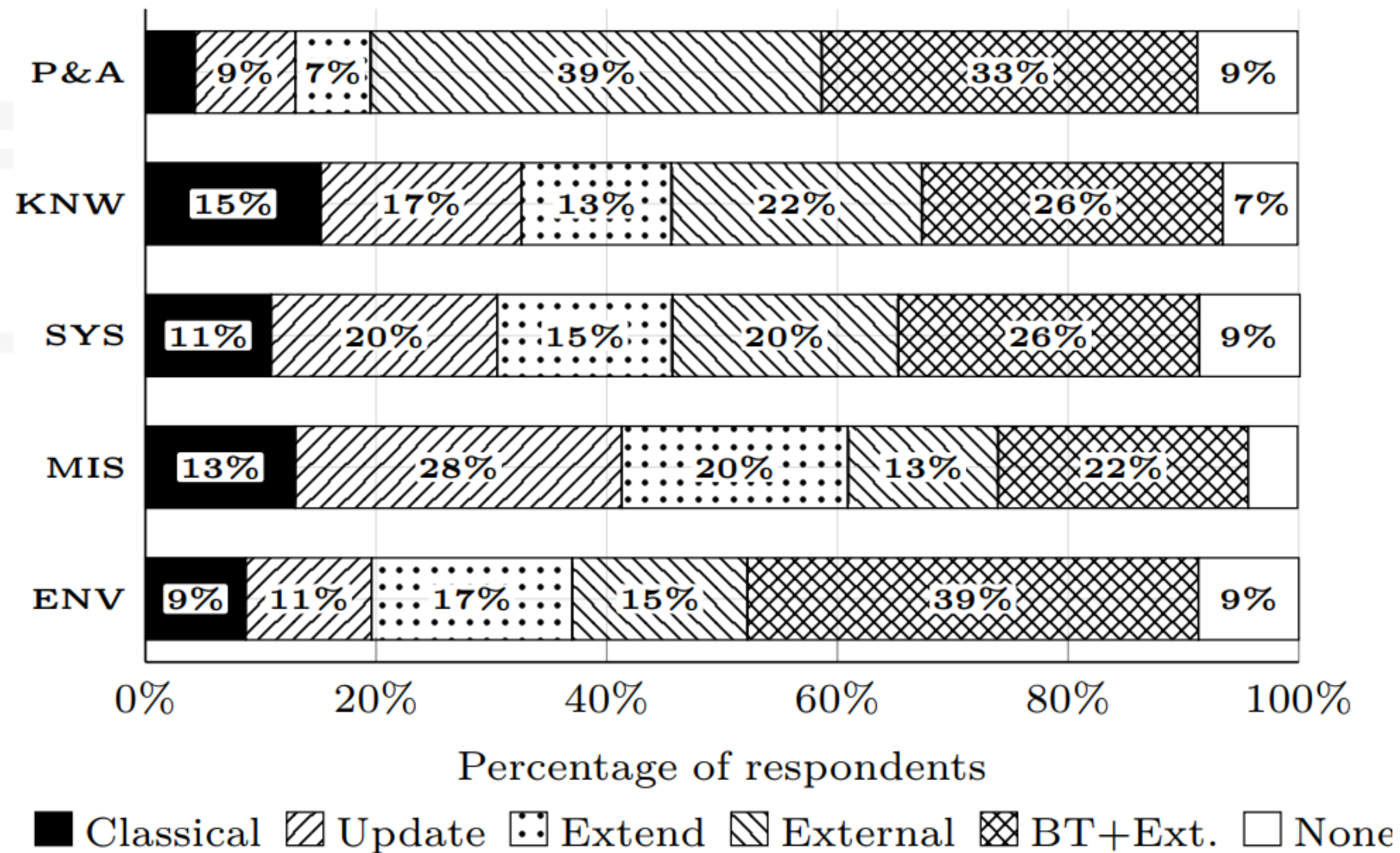
BT Refinement

BT Refinement refers to processes that take an existing, meaningful BT as input and update it to improve, correct, or adapt its behavior without changing the underlying BT formalism.



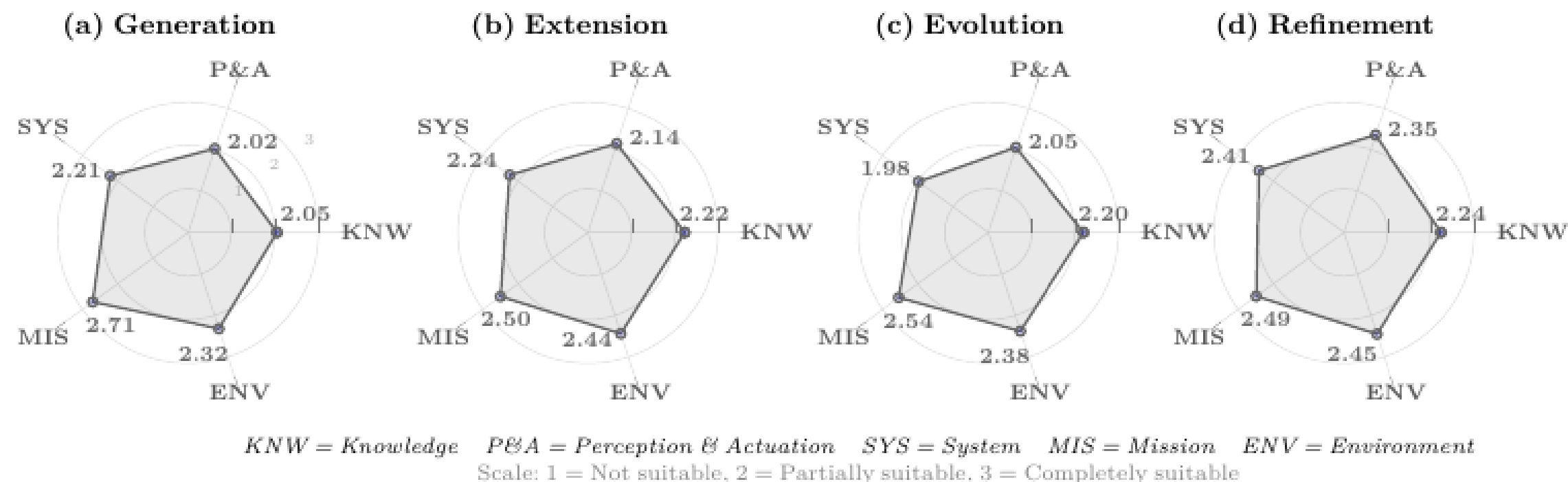
Only 4-15% saw classical BTs as sufficient

P&A = Perception & Actuation KNW = Knowledge
SYS = System MIS = Mission ENV = Environment



No enhancement family wins everywhere

Mission is strongest. Refinement is most balanced. Perception, actuation, and system needs still depend on capabilities outside the BT.



BTs are a backbone, not the adapter

The boundary is not usefulness; it is how open and uncertain the adaptation space becomes.

- 01 Strong when adaptation is bounded**

Known tasks, observable conditions, and predefined recovery strategies exploit BT modularity and reactivity.
- 02 Structural limits appear as uncertainty opens**

Mission ambiguity, resource changes, partial observability, and human coexistence require broader reasoning.
- 03 Several needs remain first-class gaps**

Perception quality, actuator loss, new resources, communication, and mission lifecycle changes remain weakly covered.

Enhanced BTs widen the space - but add cost, verification, scalability, and interpretability trade-offs.

BTs play three roles in adaptation

Calling an approach 'BT-based' does not mean the BT itself performs the adaptation.

01 The BT is the mechanism

Reactive execution selects among predefined branches when monitored conditions change.

02 The BT is the object

Refinement or evolution changes the tree, its parameters, or its execution logic.

03 The BT is the product

A planner, reasoner, learner, or LLM decides the adaptation and generates a BT for execution.

Much of the adaptation happens around the BT - while the BT remains the execution backbone.

Where should adaptation live?

- Which adaptation needs belong inside the BT - and which should remain external?
- Should runtime adaptation modify the tree, its parameters, or the surrounding architecture?
- What evidence would make a generated or refined BT safe enough to deploy?

Thank you!

 Search

Computer Science > Robotics

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Adaptation Needs in Robotic Systems: Assessing Behavior Trees and Their Enhancement

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Robotic systems increasingly operate in dynamic, uncertain, and open-ended environments, where design-time assumptions may no longer hold, and adaptation becomes necessary to maintain effective and safe operation. Behavior Trees (BTs) are widely used in robotic control architectures due to their modularity, readability, and reactivity. This raises a central question: are BTs sufficient to meet the adaptation needs of modern robotic systems?

This paper investigates this question through a literature-driven study complemented by empirical validation. First, we derive a classification of robotic adaptation needs from the literature, organizing them into six categories: Knowledge, Perception, Actuation, System, Mission, and Environment. Then, we analyze the capabilities and limitations of classical BTs with respect to these needs. Then, we characterize BT-based approaches for adaptation from the existing literature and organize them into four primary families, i.e., generation, extension, evolution, and refinement, including approaches that combine multiple families.

Our analysis shows that the modularity, flexibility, and reactivity of classical BTs are insufficient for adaptation needs involving runtime restructuring, reasoning under uncertainty, mission reinterpretation, learning, or integration with external knowledge and planning mechanisms. Enhanced BT approaches address several of these limitations, but to different extents and often with limitations of their own. Our findings relate adaptation needs to both the capabilities and limitations of classical and enhanced BTs, providing guidance on when classical BTs are sufficient, when enhanced mechanisms are needed, and which challenges remain or emerge for adaptive robotic control architectures.



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